

ENGR 102 Summer Bridge

Day 14 Student Quiz

Lists, Length, Indexes, Mutation, and List Loops

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace stored state and exact output.

4 points

```
data = [5, 2, 8, 2]
positions = []
total = 0
for index in range(len(data)):
    if data[index] == 2:
        positions.append(index)
    total = total + data[index] * index
print(total, positions)
```

Give an index ledger and exact output.

Question 2. Repair a list-bound error.

3 points

```
data = [4, 7, 9]
for index in range(len(data) + 1):
    print(data[index])
```

State the output before failure, name the failing index, and give the minimal repair.

Question 3. Program construction

13 points

Write one complete Python program that satisfies every requirement.

- Use `data = [5, 12, 5, 18, 2]`. Do not use `sum`, `max`, or `index`.
- Loop by index. Compute the sum of values stored at even indexes.
- Count all values at or above 10.
- Find the index of the first largest value.
- Print `even_index_sum`, `high_count`, and `largest_index`. Include one tie test.

[illegible]

Test evidence / expected result